Activity: Develop an AI Agent to Detect Lane Markings from Road Images Under Different Weather Conditions

Aim

To develop an AI agent that detects lane markings from road images under different weather conditions such as sunny, rainy, and foggy environments.

Problem Statement

Lane detection is an important application of Artificial Intelligence used in autonomous vehicles. The objective is to identify lane markings accurately from road images even when weather conditions affect visibility.

Methodology

1. Read the road image.
2. Convert the image to grayscale.
3. Apply Gaussian Blur to remove noise.
4. Detect edges using the Canny Edge Detector.
5. Select the Region of Interest (ROI).
6. Detect lane lines using the Hough Line Transform.
7. Draw the detected lane markings on the original image.
8. Display the output.

Algorithm

1. Start.
2. Load the road image.
3. Convert image to grayscale.
4. Apply Gaussian Blur.
5. Perform Canny edge detection.
6. Apply Region of Interest masking.
7. Detect lane lines using Hough Transform.
8. Draw detected lanes on the image.
9. Display the output image.
10. Stop.

Python Program

```
import cv2
import numpy as np

# Read image
image = cv2.imread("road.jpg")

# Convert to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Blur image
blur = cv2.GaussianBlur(gray, (5,5), 0)

# Edge detection
edges = cv2.Canny(blur, 50, 150)

# Detect lane lines
lines = cv2.HoughLinesP(edges,
                        1,
                        np.pi/180,
                        50,
                        minLineLength=100,
                        maxLineGap=50)

# Draw detected lines
if lines is not None:
    for line in lines:
        x1,y1,x2,y2 = line[0]
        cv2.line(image,(x1,y1),(x2,y2),(0,255,0),3)
```

```
cv2.imshow("Detected Lane", image)

cv2.waitKey(0)

cv2.destroyAllWindows()
```

Sample Output

Input Image:

Output:

Result

Thus, an AI agent for lane detection was successfully developed using **OpenCV**. The system detects lane markings from road images using image preprocessing, edge detection, and Hough Line Transform. It can identify lanes under different weather conditions with reasonable accuracy.